THE COHERENT SHEAVES STACK SATISFIES THE RIM-SCHLESSINGER CONDITION ON PUSHOUTS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Situation situation-coherent (canonical tag 08KB). cited at statement line 1356. Standing situation.
- Lemma more-morphisms-lemma-pushout-along-thickening (canonical tag 07RT). cited at proof line 1365. Pushouts along thickenings with affine legs.
- Lemma spaces-pushouts-lemma-pushout-along-thickening-schemes (canonical tag 07SY). cited at proof line 1381. B(Y') -> B(Y) x B(Z) x B(Z') is a bijection.
- Remark remark-coherent-base-change (canonical tag 08LP). cited at proof line 1392. Base-change compatibility of the coherent stack.
- Lemma spaces-pushouts-lemma-space-over-pushout-flat-modules (canonical tag 08KV). cited at proof line 1396. Modules over the pushout give the equivalence.

Source reference. Original definition 08KB: sources/stacks-chow/tags.txt, source lines 10881–10881. Cited at statement line 1356: standing situation.

Source reference. Original lemma 07RT: sources/stacks-chow/tags.txt, source lines 9878–9878. Cited at proof line 1365: the pushout square.

Source reference. Original lemma 07SY: sources/stacks-chow/tags.txt, source lines 9918–9918. Cited at proof line 1381: base bijection.

Source reference. Original remark 08LP: sources/stacks-chow/tags.txt, source lines 10927–10927. Cited at proof line 1392: replace B by Y'.

Source reference. Original lemma 08KV: sources/stacks-chow/tags.txt, source lines 10898–10898. Cited at proof line 1396: the equivalence.

Lemma 0.2. *In Situation 08KB. Let*

$$\begin{array}{ccc} Z & \longrightarrow & Z' \\ \downarrow & & \downarrow \\ Y & \longrightarrow & Y' \end{array}$$

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be a pushout in the category of schemes over S where $Z \to Z'$ is a thickening and $Z \to Y$ is affine, see More on Morphisms, Lemma 07RT. Then the functor on fibre categories

$$\mathcal{C}oh_{X/B,Y'} \longrightarrow \mathcal{C}oh_{X/B,Y} \times_{\mathcal{C}oh_{X/B,Z}} \mathcal{C}oh_{X/B,Z'}$$

is an equivalence.

Proof. Observe that the corresponding map

$$B(Y') \longrightarrow B(Y) \times_{B(Z)} B(Z')$$

is a bijection, see Pushouts of Spaces, Lemma 07SY. Thus using the commutative diagram

$$\begin{array}{ccc} \mathcal{C}oh_{X/B,Y'} & \longrightarrow & \mathcal{C}oh_{X/B,Y} \times_{\mathcal{C}oh_{X/B,Z}} \mathcal{C}oh_{X/B,Z'} \\ \downarrow & & \downarrow \\ B(Y') & \longrightarrow & B(Y) \times_{B(Z)} B(Z') \end{array}$$

we see that we may assume that Y' is a scheme over B'. By Remark 08LP we may replace B by Y' and X by $X \times_B Y'$. Thus we may assume $B = Y'$. In this case the statement follows from Pushouts of Spaces, Lemma 08KV. □